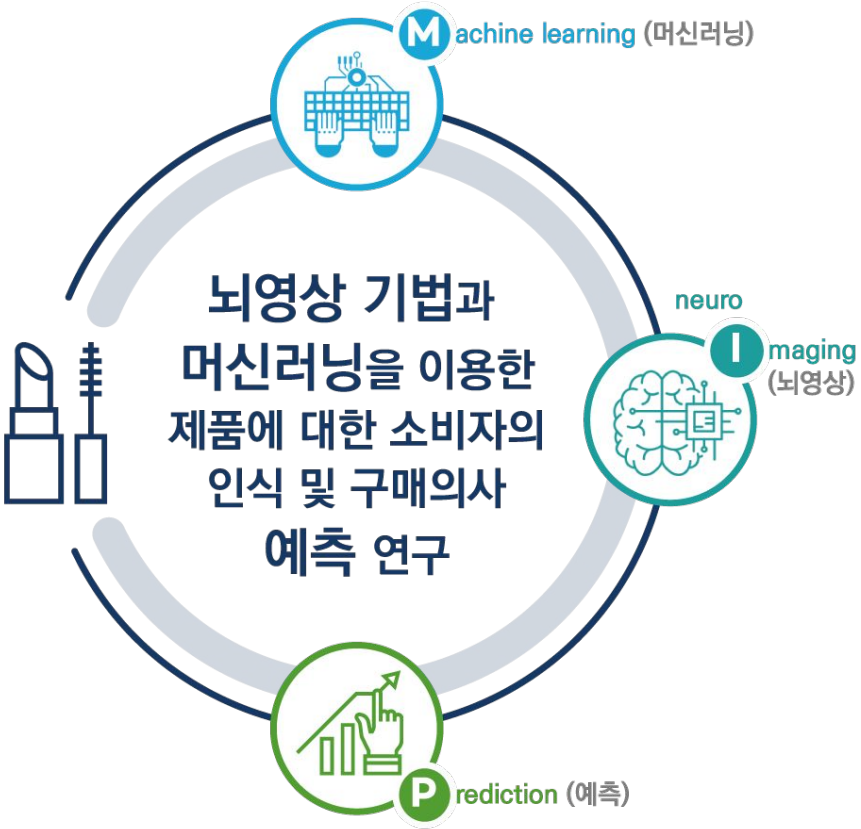

Project COSMAX

2021.12.06

Mina Kwon

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JeungHyun Lee

Graduate student (Clinical)

Saeyeon (Ella) Rho

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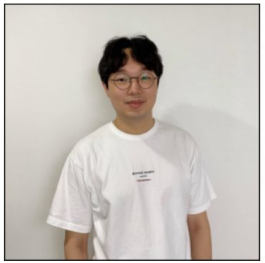
Alumni

Jooyeon (Jamie) Im

Postdoctoral Fellow



Youngha Jo

Undergraduate student

Contents

- Introduction
  - Background
  - Research Goal & Hypotheses
- Study Design
  - Participants
  - Stimuli
  - Task Design
- General Linear Model (GLM)
- Prediction using penalized logistic regression
- Future plans

Research Background (A Quick Recap)

Neuromarketing
: Marketing technique that analyses the neurological activity of the brain in order to predict consumer choices

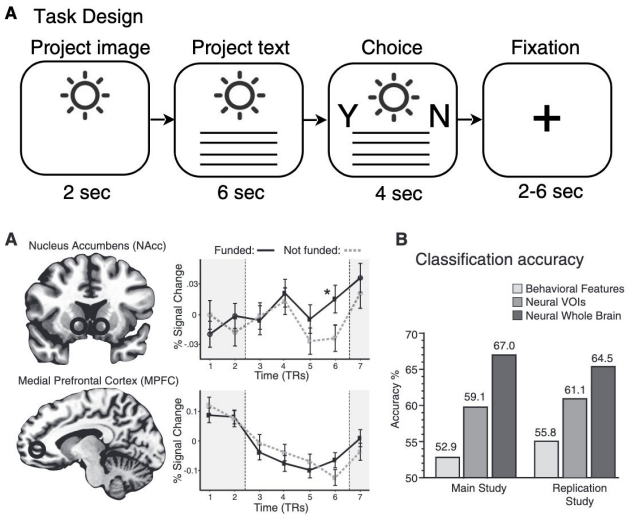
Prediction

Predicting behaviour within individuals

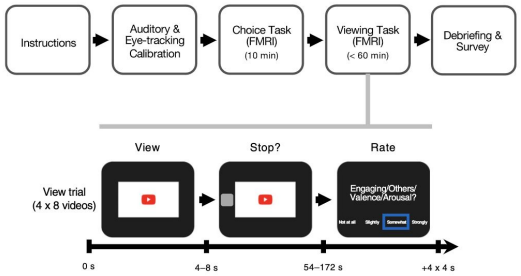
Forecast

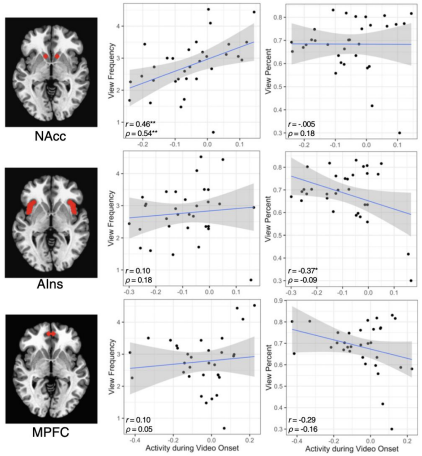
Predicting behaviour of population

Neuromarketing studies

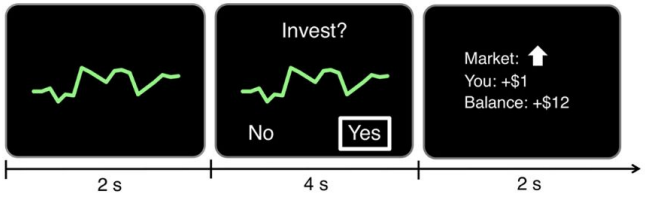


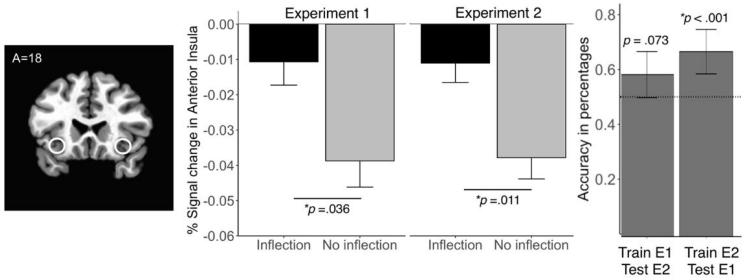
Genevsky et al. (2017), JNeuro





Tong et al. (2020), PNAS





Stallen et al. (2021), JNeuro

Literature Gap & Research Aims

1. Neuromarketing research in the Cosmetic industry is still in the early stages due to a range of variables in consumer choice

⇒ *Apply various neuroimaging techniques in predicting consumer behaviour for cosmetic products*

2. Limitations of sale rate indicators in existing forecasting studies (e.g., measurement of how much is sold in a single supermarket over a period of time; Kuhn et al., 2016)

⇒ *Use reliable and valid predictors by leveraging data from COSMAX (e.g. order quantity)*

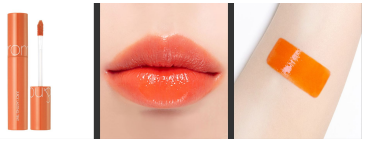
Research Aims

1. Predict individuals' preference and aggregate sales data using fMRI data

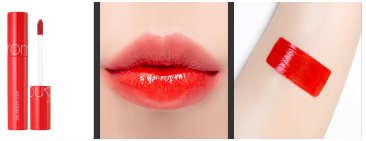
2. Examine whether neural data could better predict consumer behavior compared to survey data (Not yet)

Stimuli

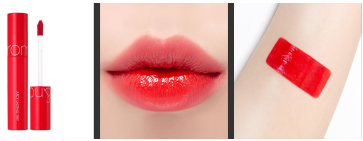
Product 1



Product 2

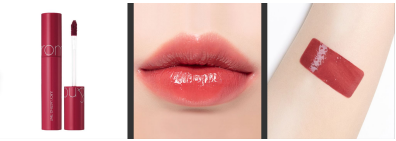


Product 3

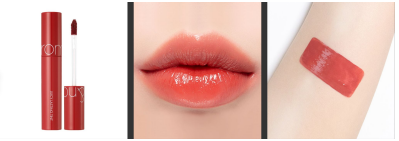


Product 4

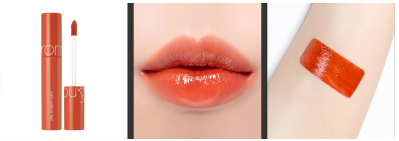
Product 6



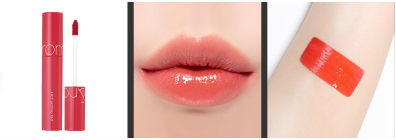
Product 7



Product 8



Product 9



Rank	Product #	Cumulative Order Quantity
1	Product 7	1667490
2	Product 6	1500319
3	Product 8	974169
4	Product 9	465547
5	Product 3	158540
6	Product 2	146392
7	Product 5	131455
8	Product 1	100835

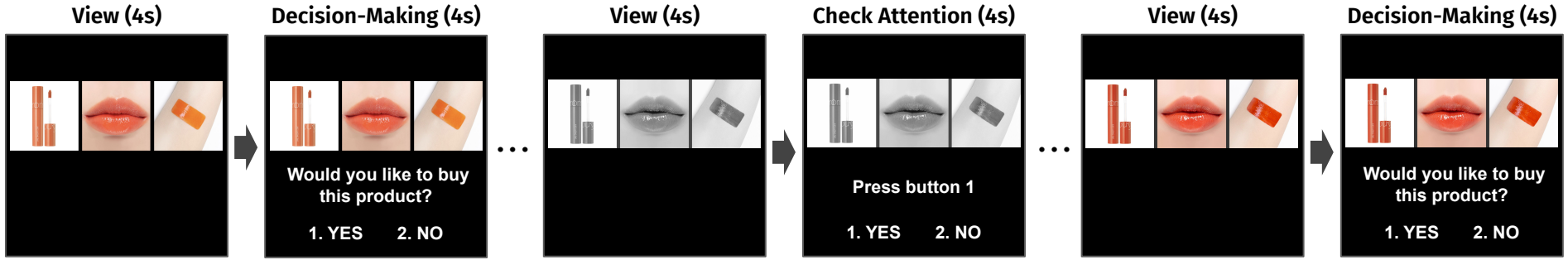
Sales data (Order Quantity)

Study Design

Participants

54 (out of 60; all females; Mean age = 22.72 (sd = 3.59)

1 session (6 sessions in total)



A decision-making trial
(1 product)

An attention trial
(black/white stimuli, 1 trial per session)

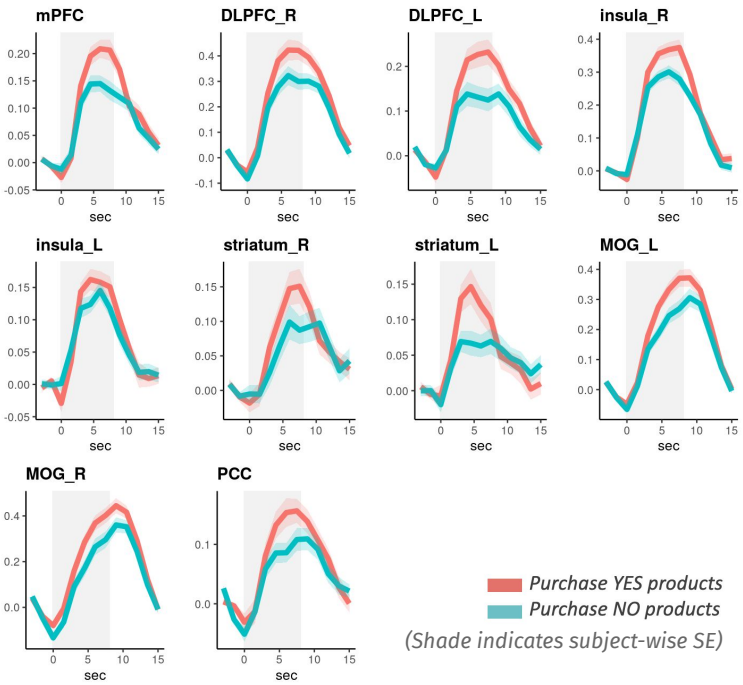
A decision-making trial
(1 product)

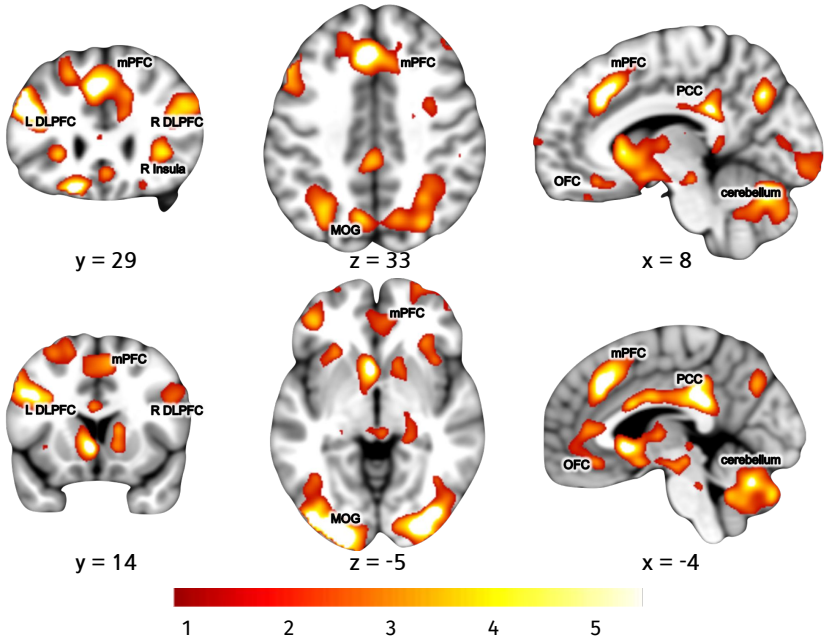
GLM (whole-brain analysis)

GLM 1. Prediction of individual preferencee

1. Prediction (***Purchase YES products - Purchase NO products, based on task response***)

N = 54, P < 0.05 (FDR corrected)





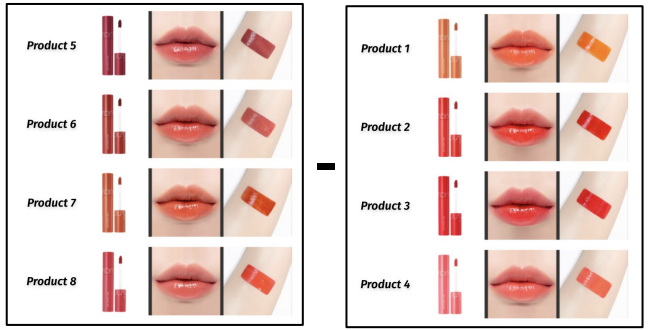
Time-series signals of the peak voxel of each regions (radius 3mm)

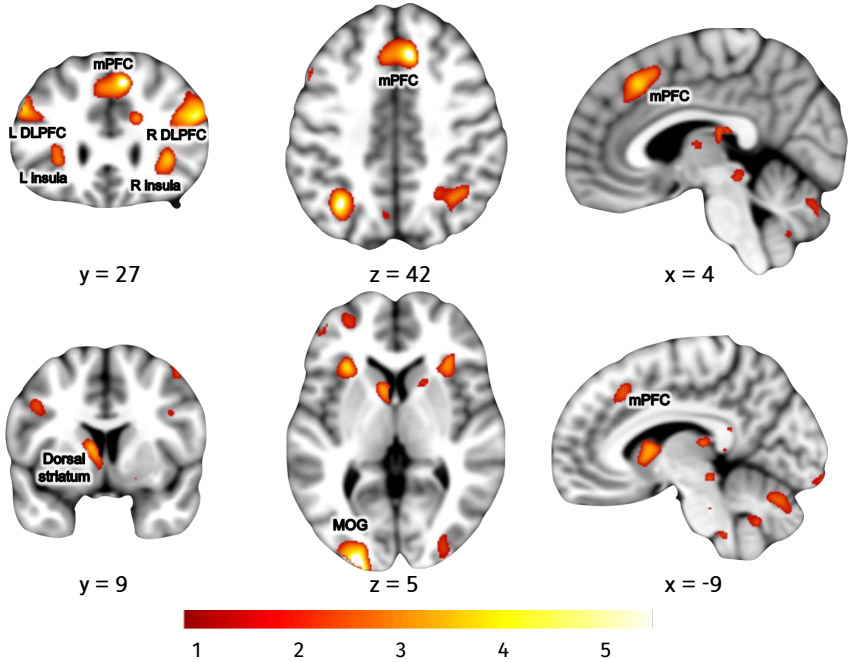
GLM 2. Forecast of sales (popularity)

2. Forecast (***Popular products - Unpopular products***)

N = 54, P < 0.05 (FDR corrected)

Brain activity while viewing products with high sales
Vs.
while viewinng products with low sales





Trial-by-trial Prediction:
Which product is being presented?

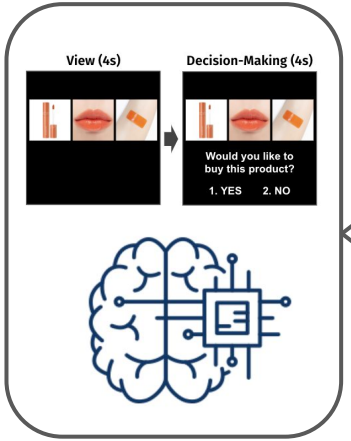
Two types of prediction

Prediction

Predicting behaviour within individuals

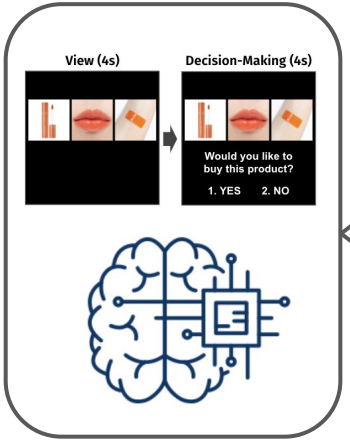
Forecast

Predicting behaviour of population



Products with >50% YES purchase

Products with >50% NO purchase

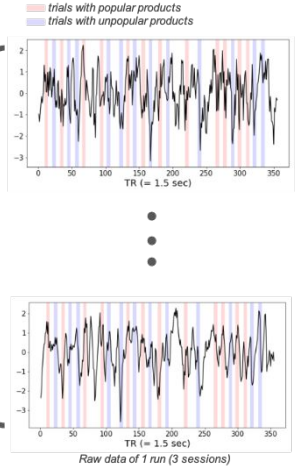


Popular products with High sales

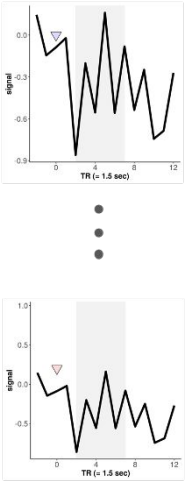
Unpopular products with Low sales

Method - Analysis

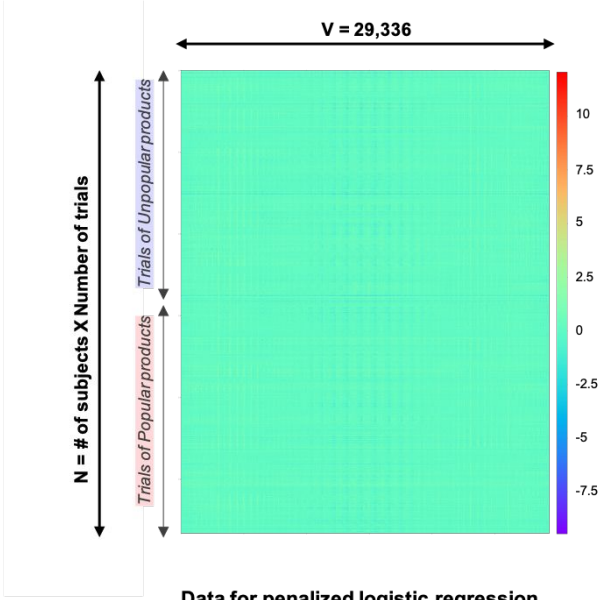
A priori mask
(reward + preference)



Extract raw data from each voxel
(29,336 voxels in the mask)



Average signals of each trial for each voxel
(48 trials)



Data for penalized logistic regression

Method - Analysis

CCS-Lab / ccsl_fmri_pipeline Private

Watch 4 | Star 1 | Fork 0

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights

main | ccsl_fmri_pipeline / nilearn_codes / time_series / | Go to file | Add file

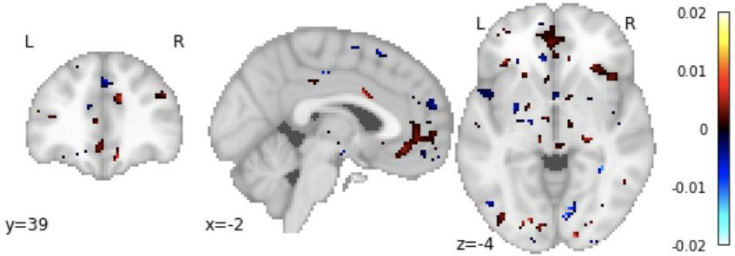
NeuroVR-Song minor codes change 38c1a25 on Sep 8 History

..		
.ipynb_checkpoints	minor codes change	3 months ago
__pycache__	minor codes change	3 months ago
ni_ts	codes directory change	3 months ago
spm_ts	codes directory change	3 months ago
L_amyg-interp.nii	fix error fmriprep mask img	4 months ago
ROI_mask_validation.ipynb	fix error fmriprep mask img	4 months ago
R_amyg-interp.nii	fix error fmriprep mask img	4 months ago
example_time_series.ipynb	minor codes change	3 months ago
explanation.ipynb	time series code update	3 months ago
extract_rois.py	time series code update	3 months ago
nilearn_sub-001_ses-01_run-01_timeseries.csv	fix error fmriprep mask img	4 months ago
sample.txt	fix error fmriprep mask img	4 months ago
time_series.py	minor codes change	3 months ago
time_series_validation_N10_1.ipynb	fix error fmriprep mask img	4 months ago
time_series_validation_N10_2.ipynb	fix error fmriprep mask img	4 months ago
time_series_validation_N10_3.ipynb	fix error fmriprep mask img	4 months ago
timeseries.csv	fix error fmriprep mask img	4 months ago

Myeong Seop Song
Postdoctoral Fellow

Preliminary results

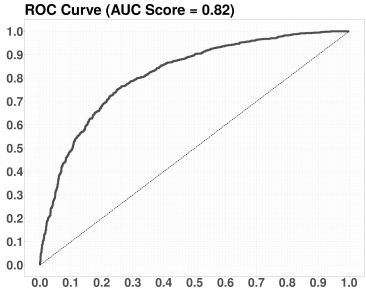
Prediction, *3,082 voxels survived*



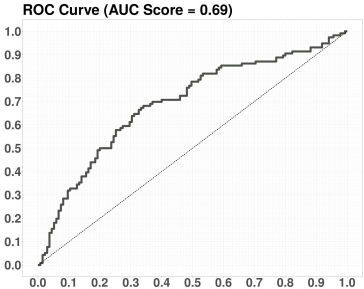
Coefficients of survived voxels

Penalized logistic regression

- Libraries: Easyml & scikit-learn
- Elastic net (α = 0.01)
- n_divisions = 10, train_size = 0.9
- n_samples = 10, n_iterations = 10



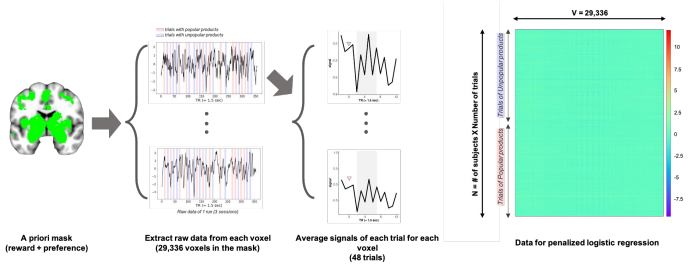
Training set



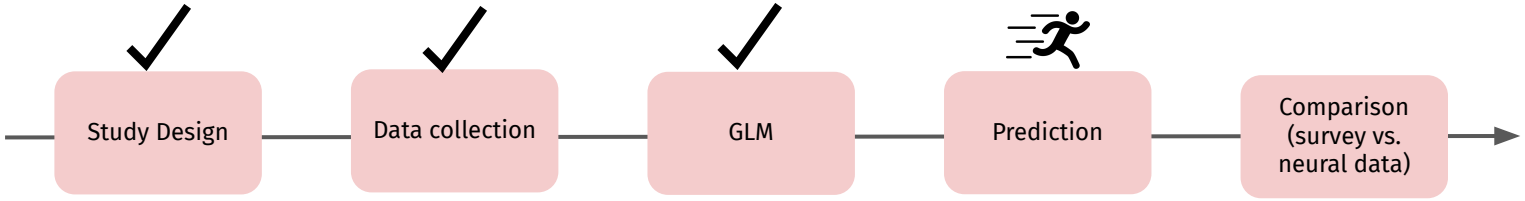
Test set

Future plans

- Use only the first exposure (Ahn et al., 2014)
- Use other measure of temporal reduction
- Compare prediction accuracy of survey vs. survey + neural data



Model	AUC score
Predict Purchase response ~ preference + expected popularity	0.835
Forecast sales ~ purchase response + preference + expected popularity	0.611



Thank you :)